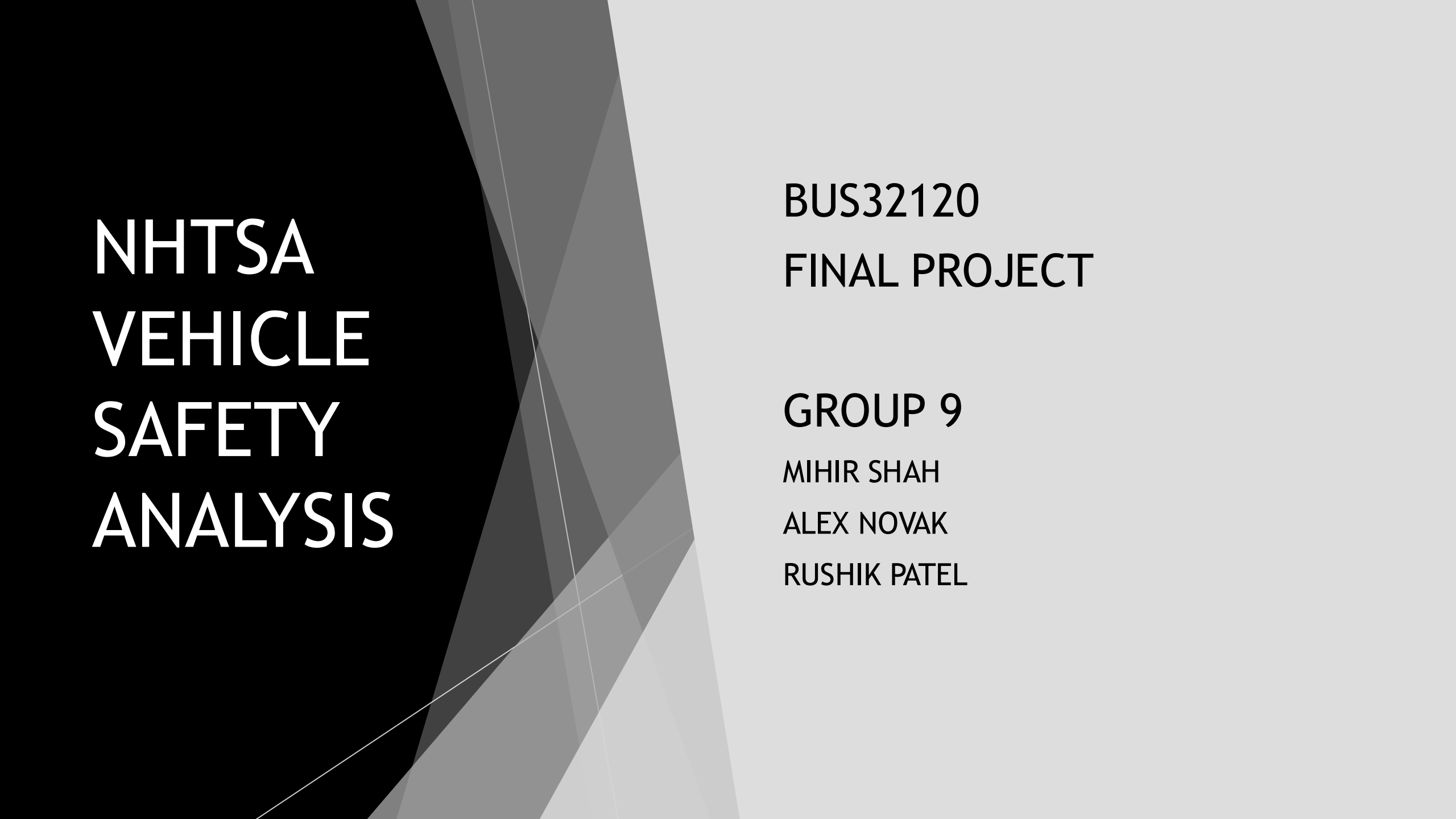




NHTSA VEHICLE SAFETY ANALYSIS

BUS32120
FINAL PROJECT

GROUP 9

MIHIR SHAH

ALEX NOVAK

RUSHIK PATEL



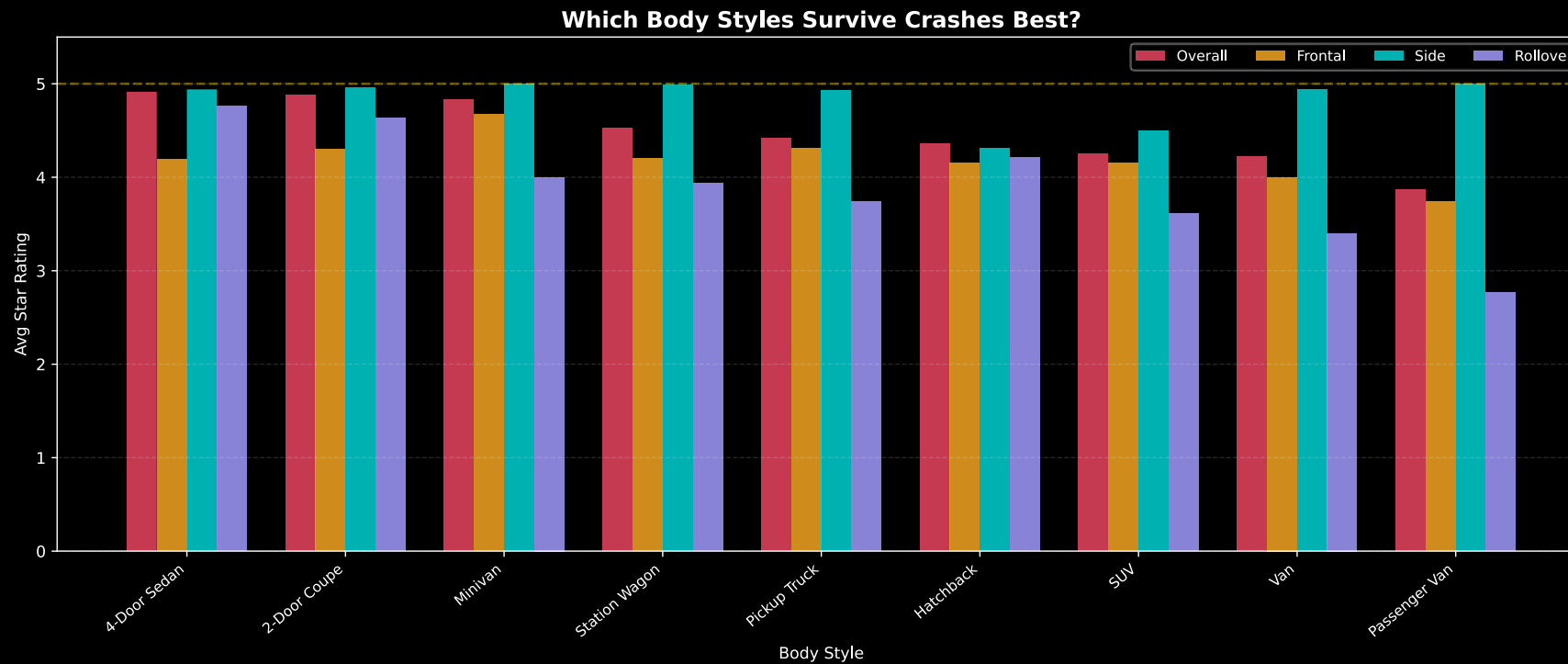
PART 1



ARE CARS ACTUALLY SAFE?

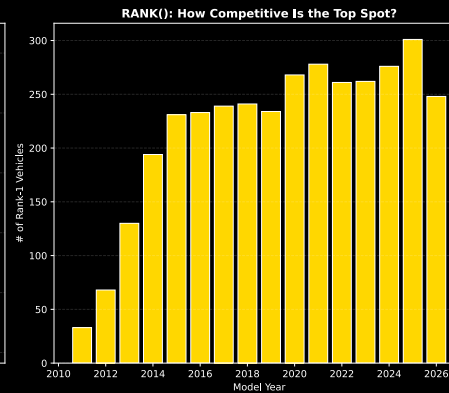
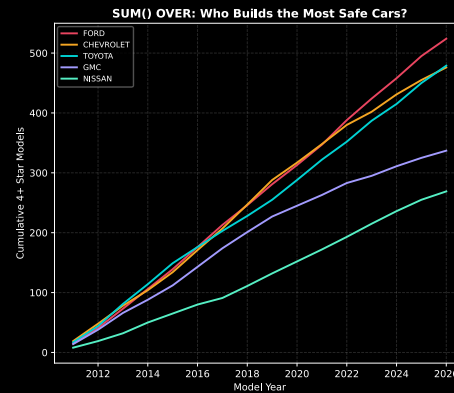
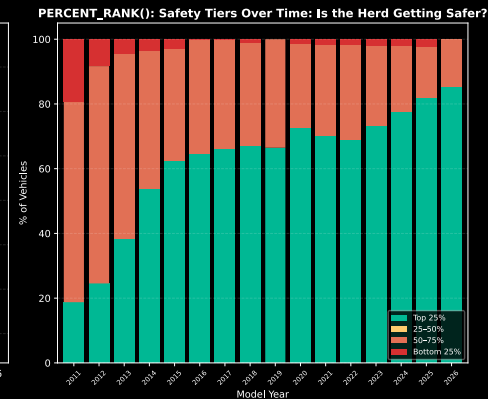
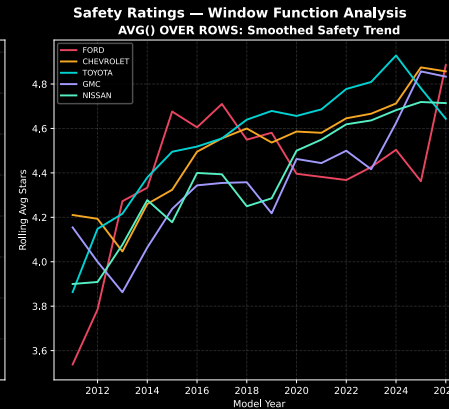
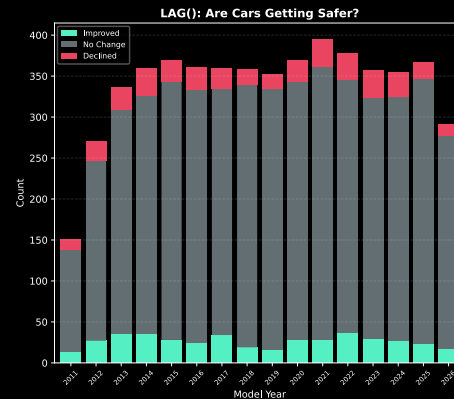
SAFETY RATINGS BY BODY STYLE

- 4-Door Sedans lead with overall star averages
- Rollover ratings show the most variation across body styles
- No body style averages a perfect 5-star score across all categories



SAFETY RATING TRENDS YEAR-BY-YEAR

- Cars are not getting safer or worse over time
- Top 5 makes hover above 4.3 stars
- Ford has the most 4+ star models in raw numbers, mostly because they test more vehicles





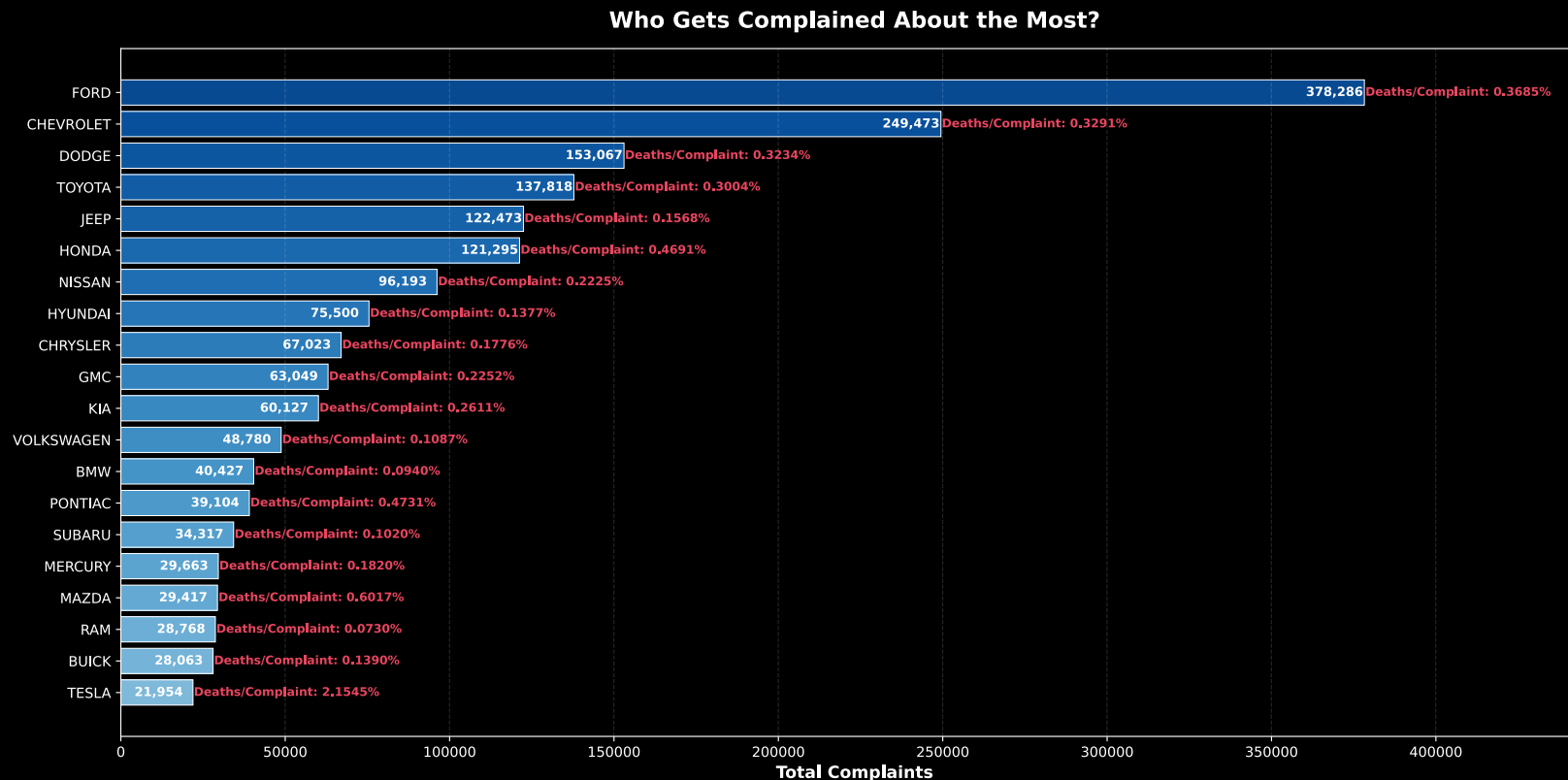
PART 2



WHAT DO CONSUMERS REPORT?

RANKING MAKES BY CONSUMER COMPLAINTS

- Ford, Chevy, and Dodge have the most complaints
- However, death rate per complaint is generally higher in mid-volume brands like Tesla



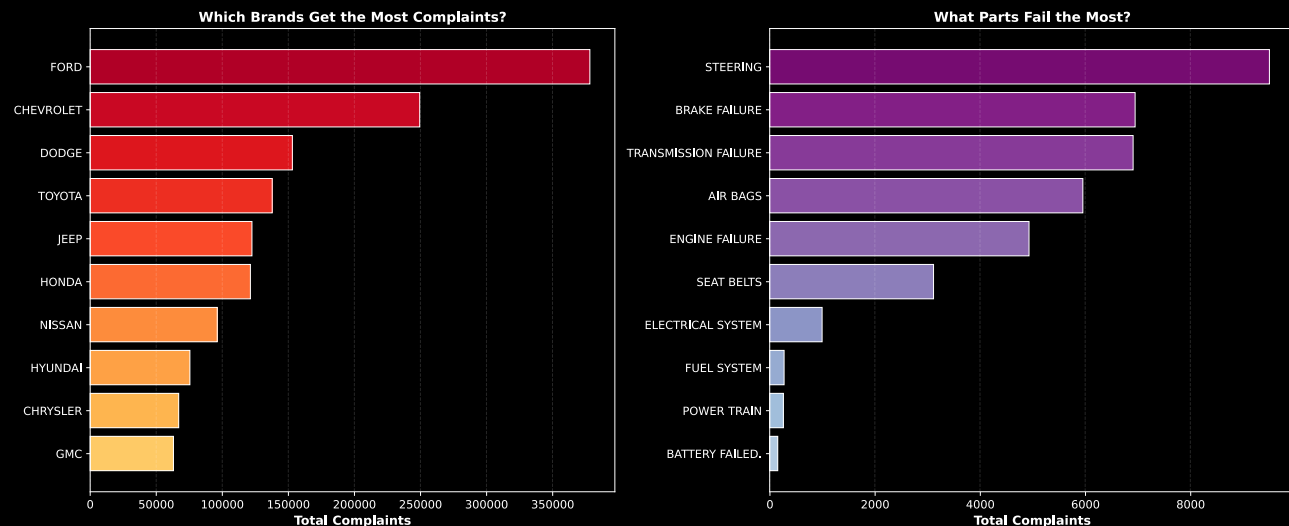
WHAT BREAKS & WHO MAKES IT?

The Complaint Leaderboard Over Time

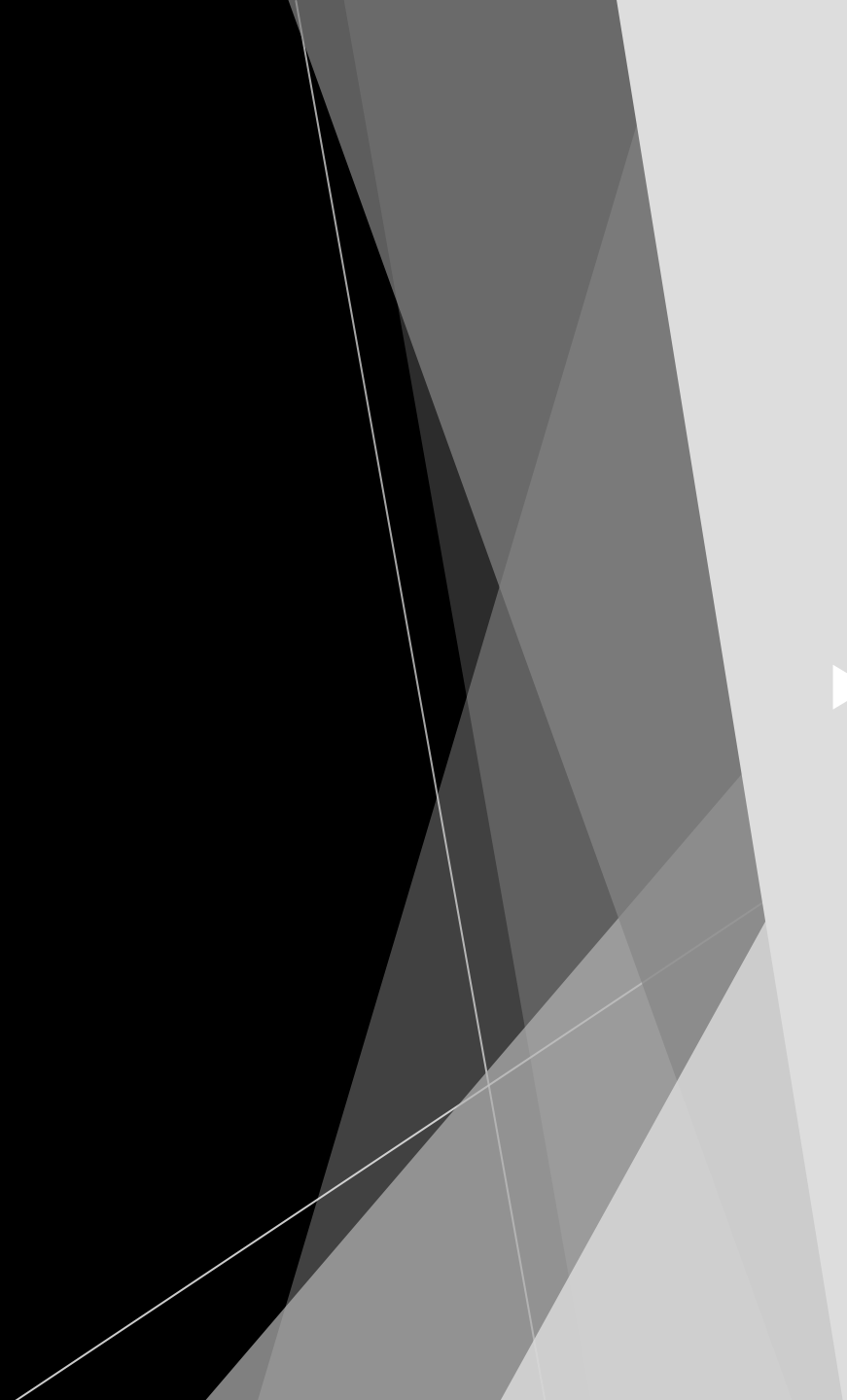


- Ford, Chevy, and Dodge have the most complaints
- Mostly due to volume of cars on the road

What Breaks and Who Makes It?



- Steering, brakes, and transmissions have the most failures

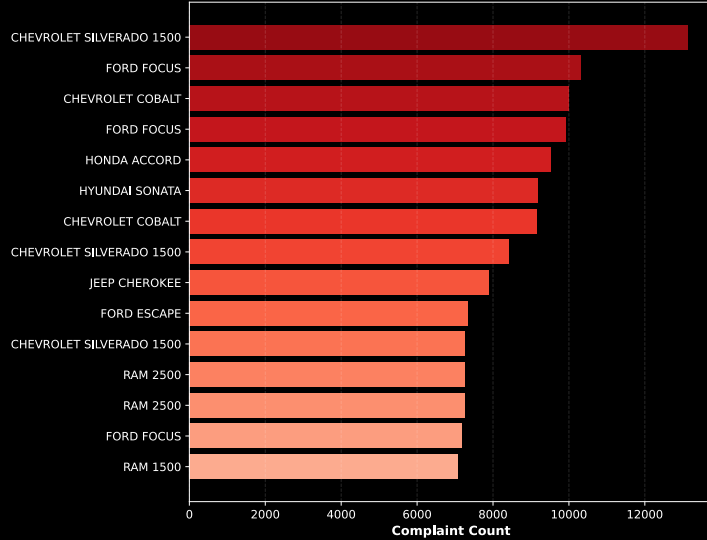


PART 3

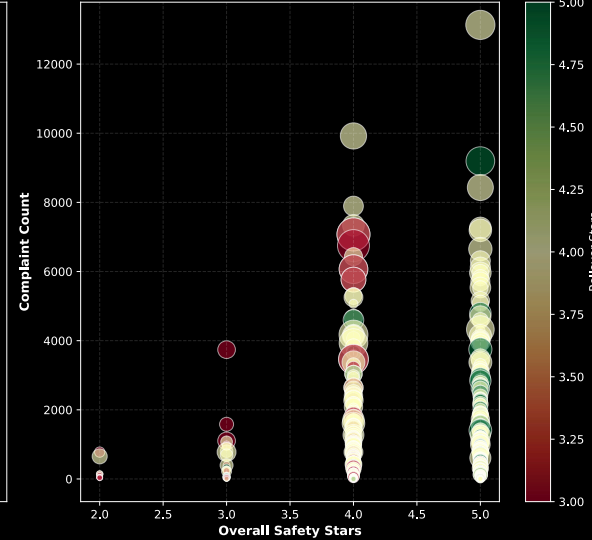
- ▶ DO COMPLAINTS MATCH SAFETY RATINGS?

DO COMPLAINTS MATCH SAFETY RATINGS? CAN WE PREDICT?

Most Complained-About Models (with Crash Ratings)



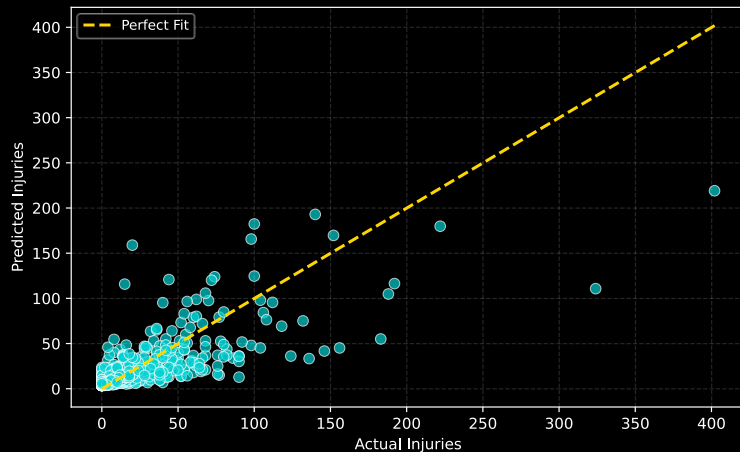
Do Safer Cars Get Fewer Complaints?



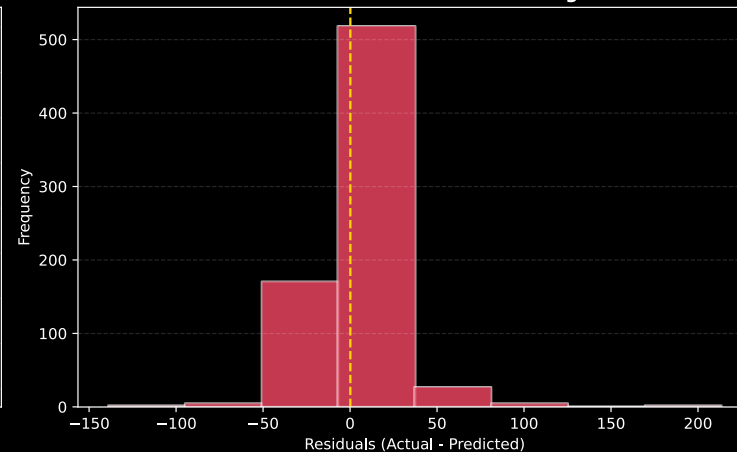
- Safer cars do not necessarily lead to fewer complaints
- Bubble size (injuries) indicate that more dangerous vehicles cluster at lower star ratings
- Vehicles with lower rollover stars show some high-complaint models have poor rollover resistance despite good frontal/side crash scores
- The regression does well on low injury cases but is not as accurate at the extreme level
- Since most models have safety stars around 4-5, it does not have much predictive impact

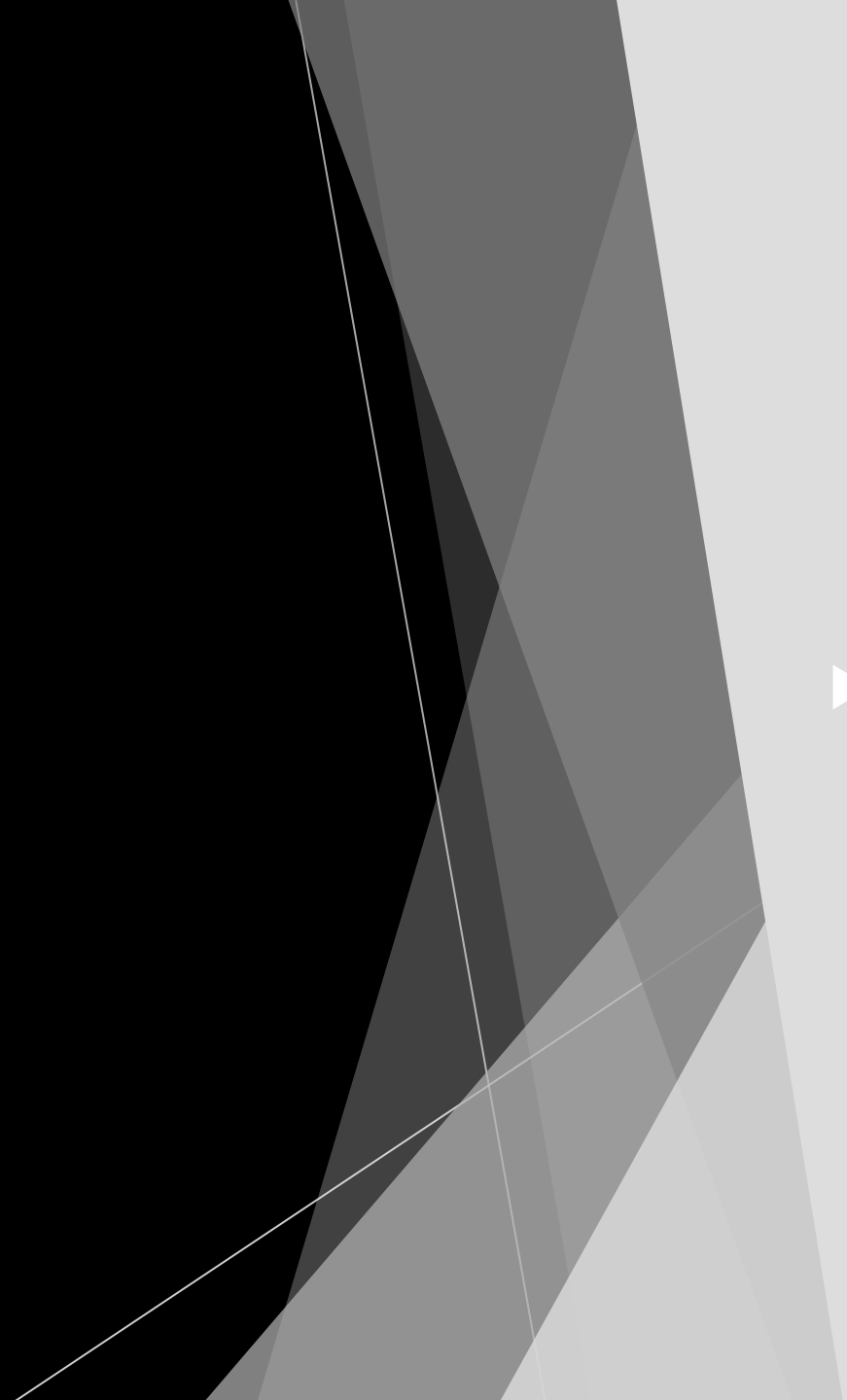
Can We Predict Injuries from Complaints & Safety Stars?

How Close Are Our Predictions?



Where Does the Model Go Wrong?



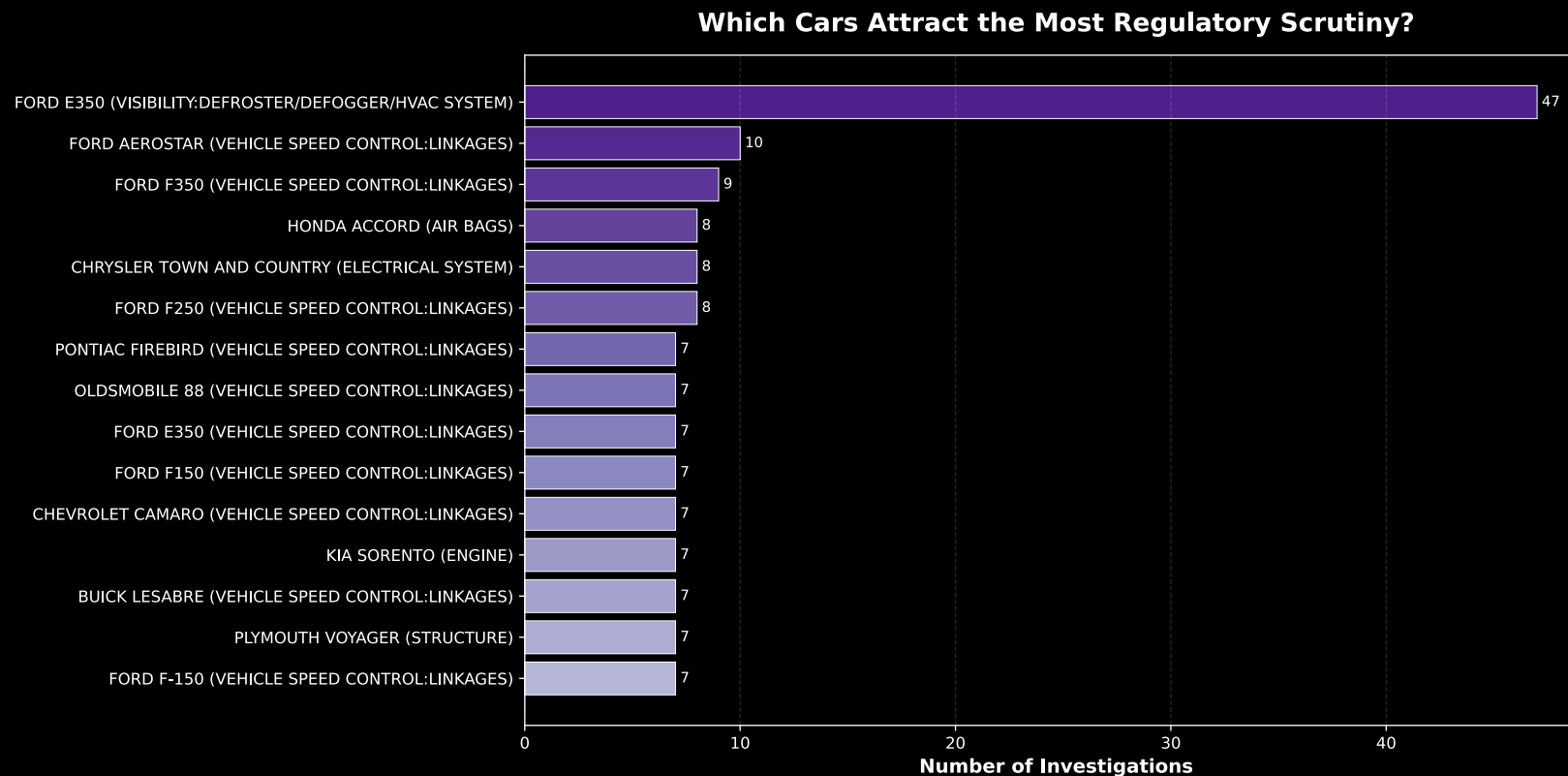


PART 4

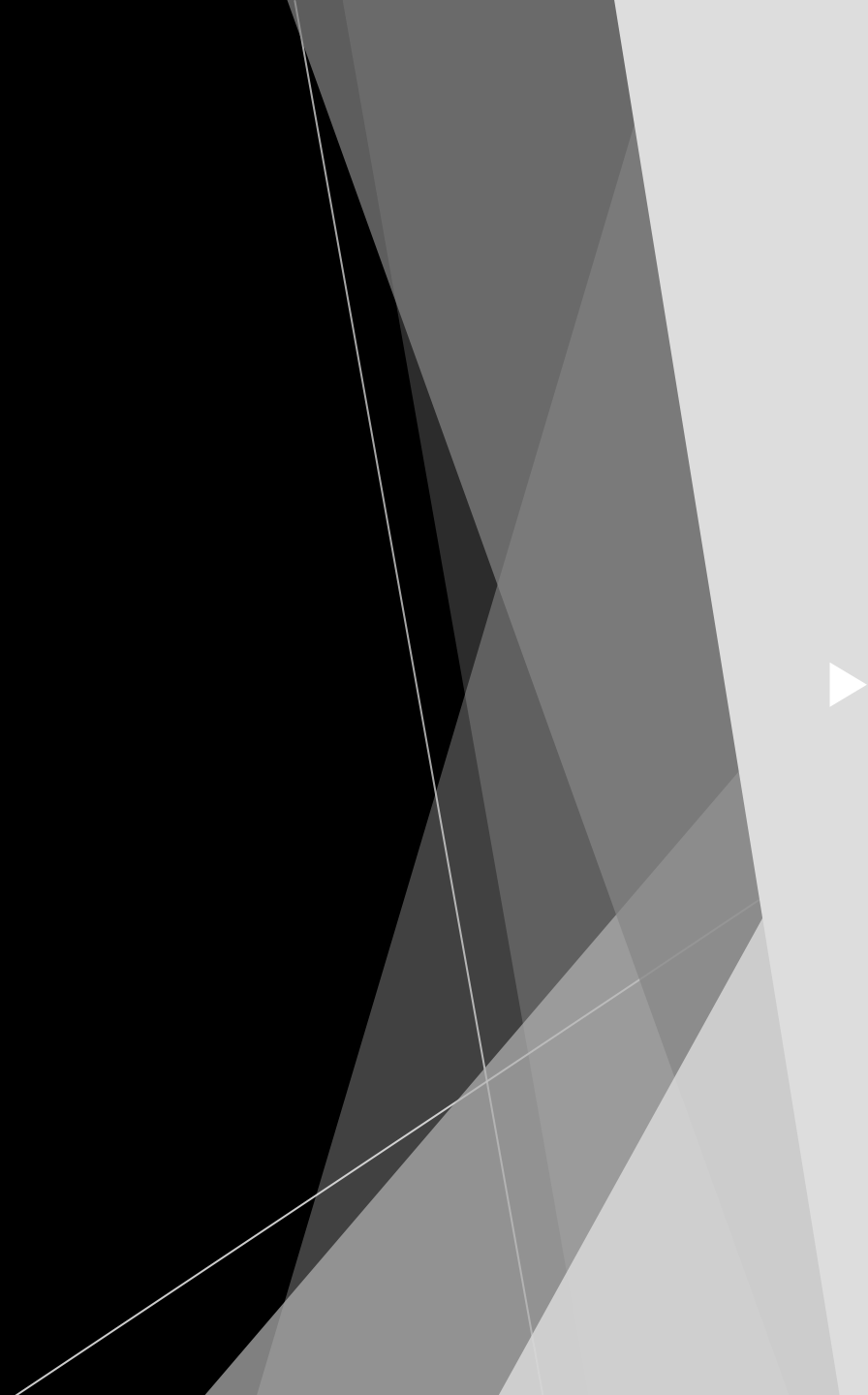


WHEN DOES NHTSA TAKE NOTICE?

WHICH CARS ATTRACT REGULATORY SCRUTINY?



- Repeated investigations into the same make/model/component shows persistent defects despite regulatory scrutiny



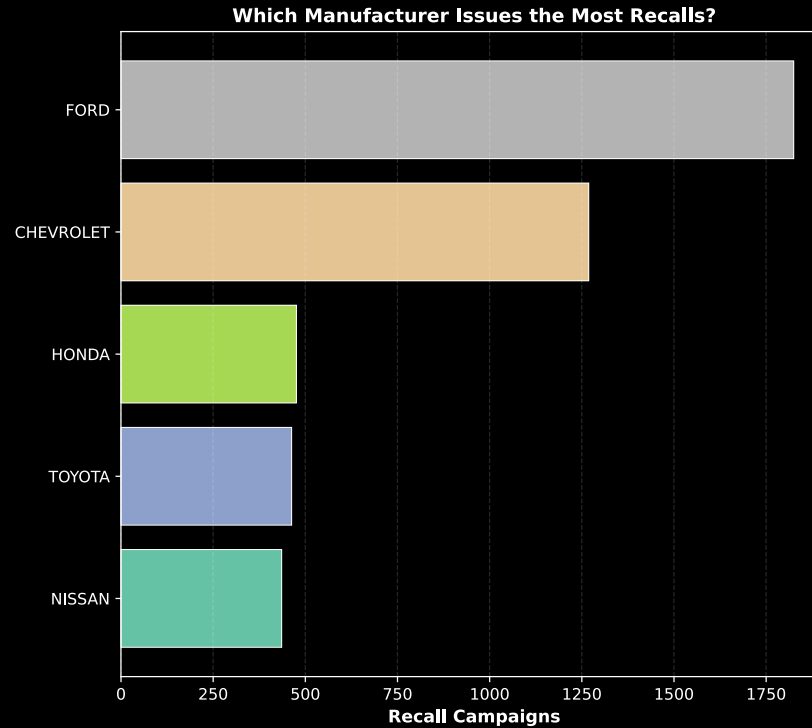
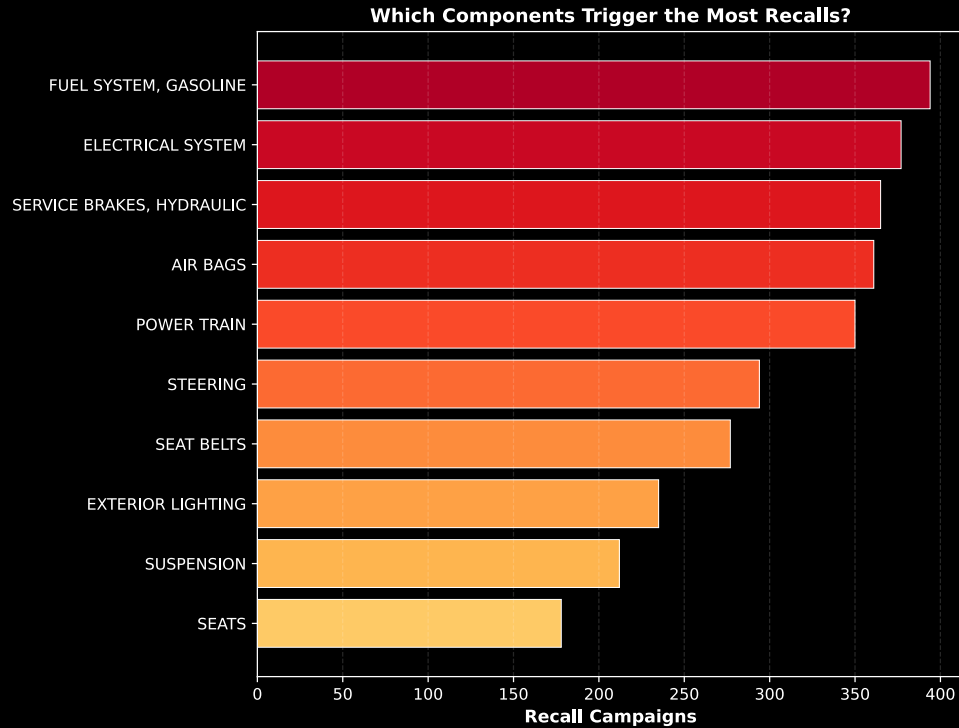
PART 5

▶ WHAT GETS RECALLED?

RECALL ANATOMY BY COMPONENT/MAKE

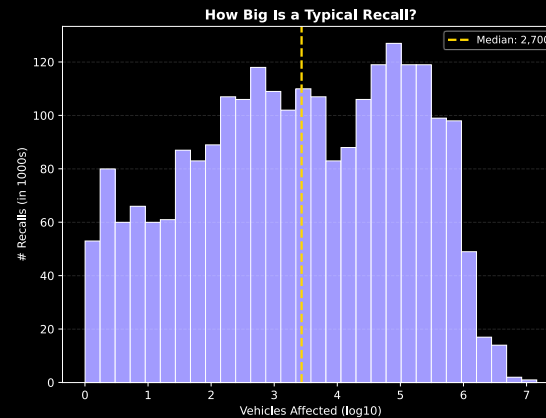
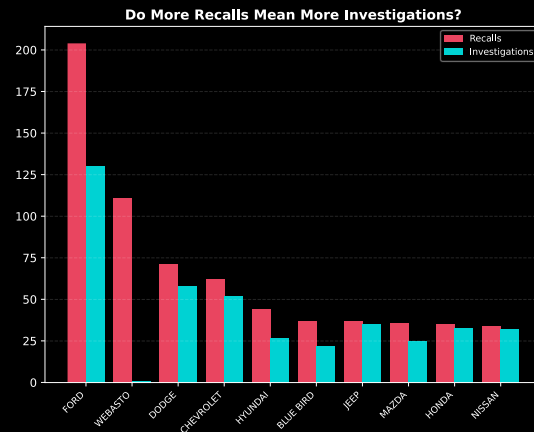
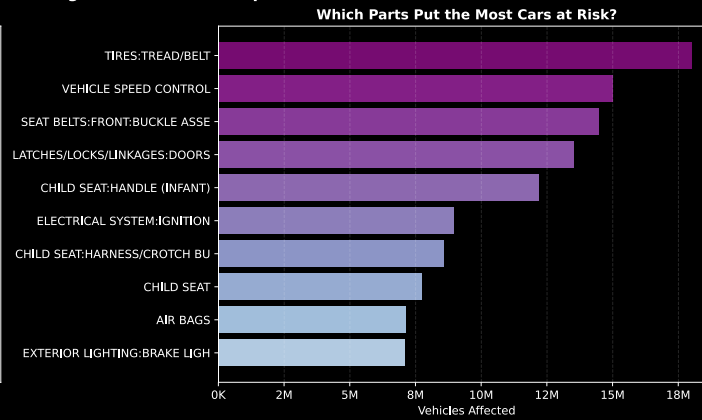
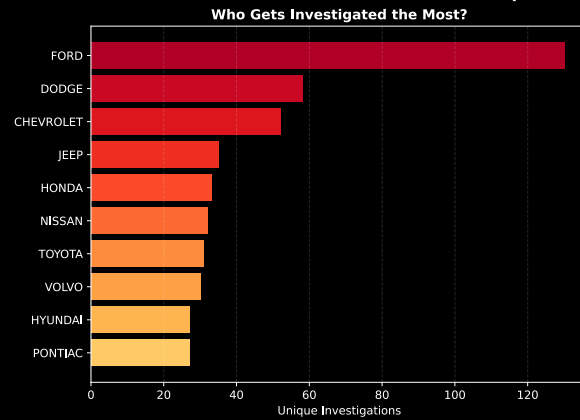
- Fuel and electrical systems trigger the most recalls
- Ford and Chevy issue the most recalls by count, consistent with their overall market share and range of affected components

The Recall Anatomy: Components vs Manufacturers

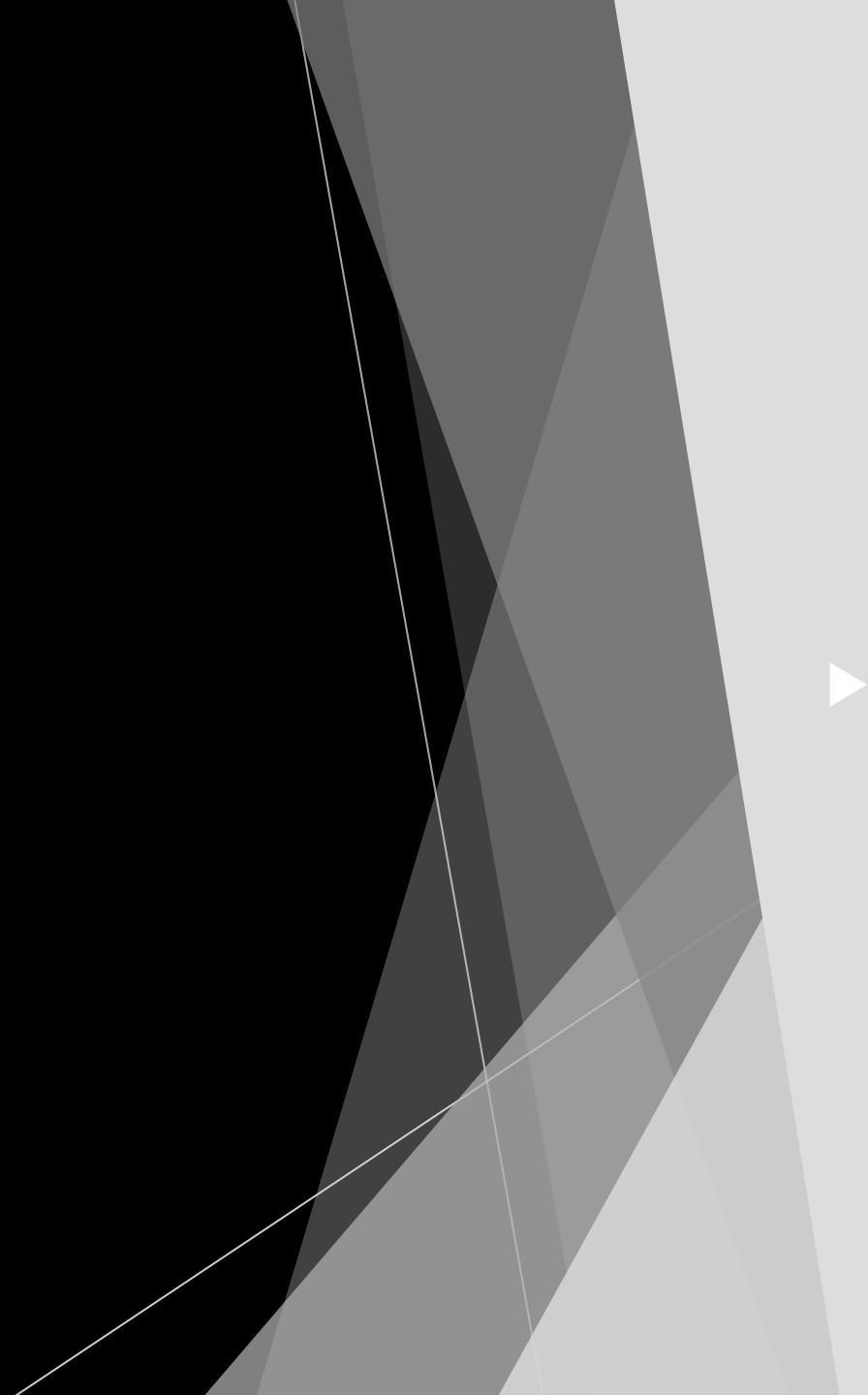


LINKING RECALLS AND INVESTIGATIONS

From Complaint to Investigation: The Recall Pipeline



- Ford, Dodge, and Chevy have the most formal investigations
- Tires and speed control drive the most recalls
- Generally more recalls = more investigations
- The log-scale histogram reveals most investigations affect 10k-100k vehicles

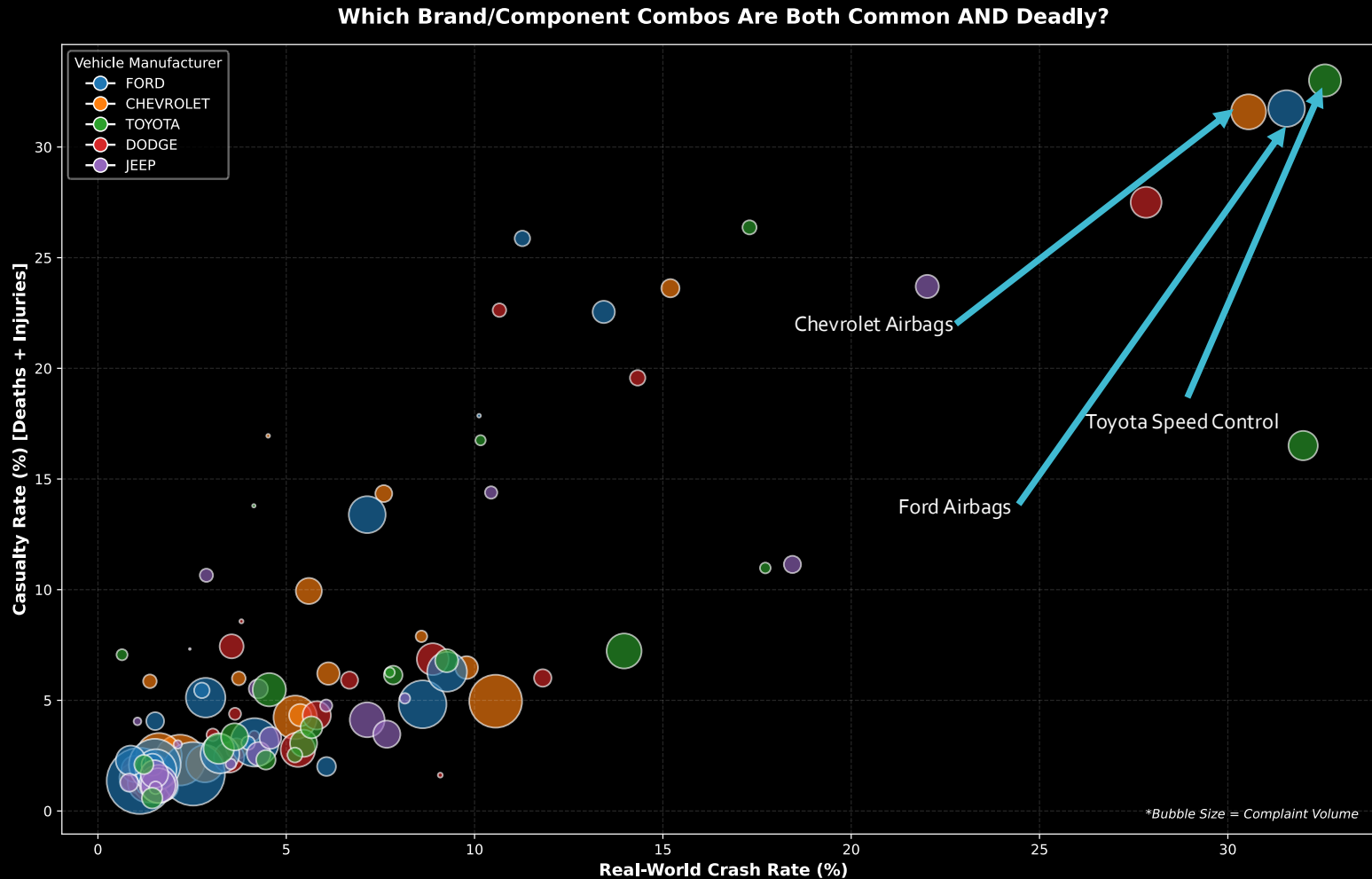


PART 6



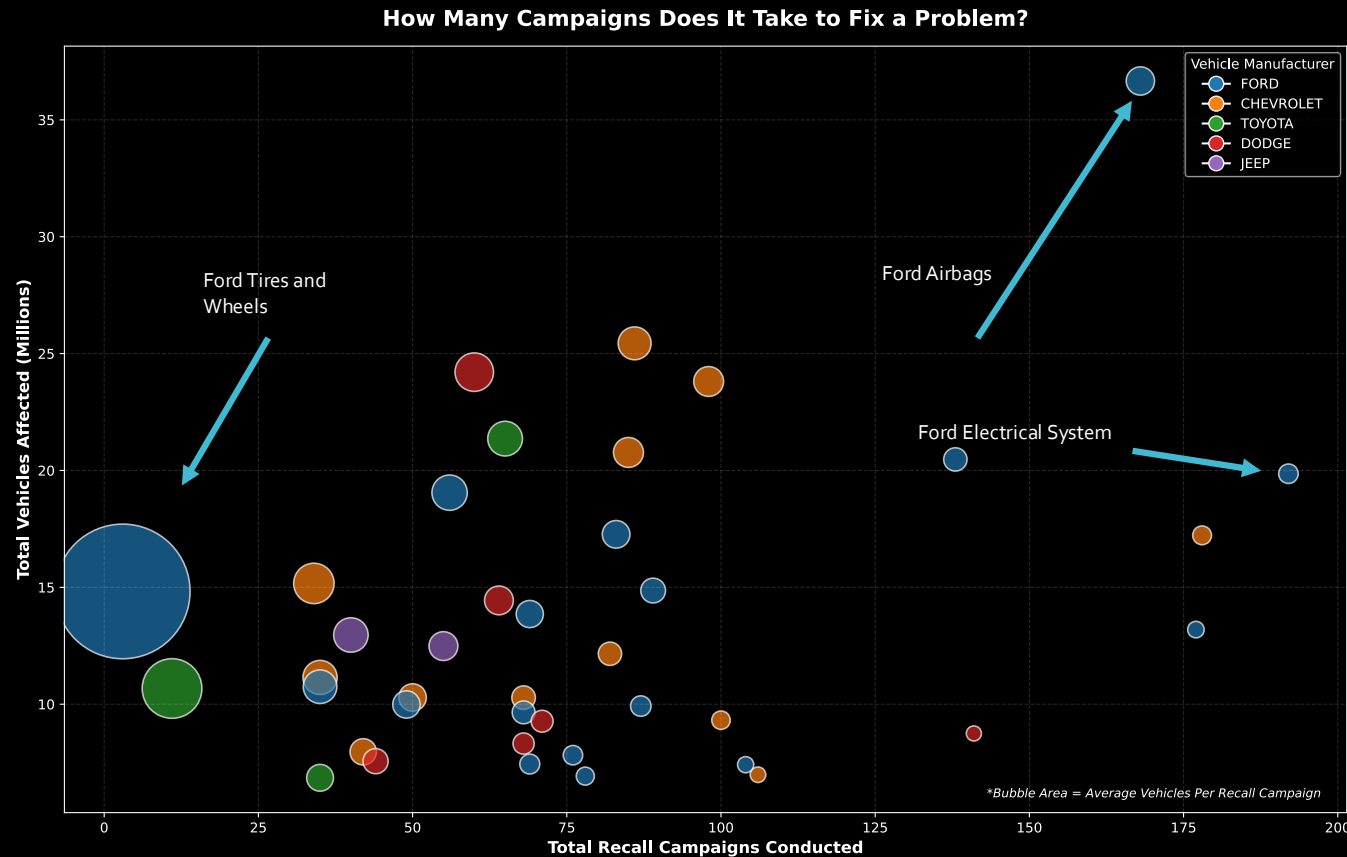
COMPLAINT → RECALL

STRATEGIC SAFETY MATRIX - CRASH RATE VS. CASUALTY RATE



- While airbags have higher crash/injury rates than most components, they are only used when a crash has occurred

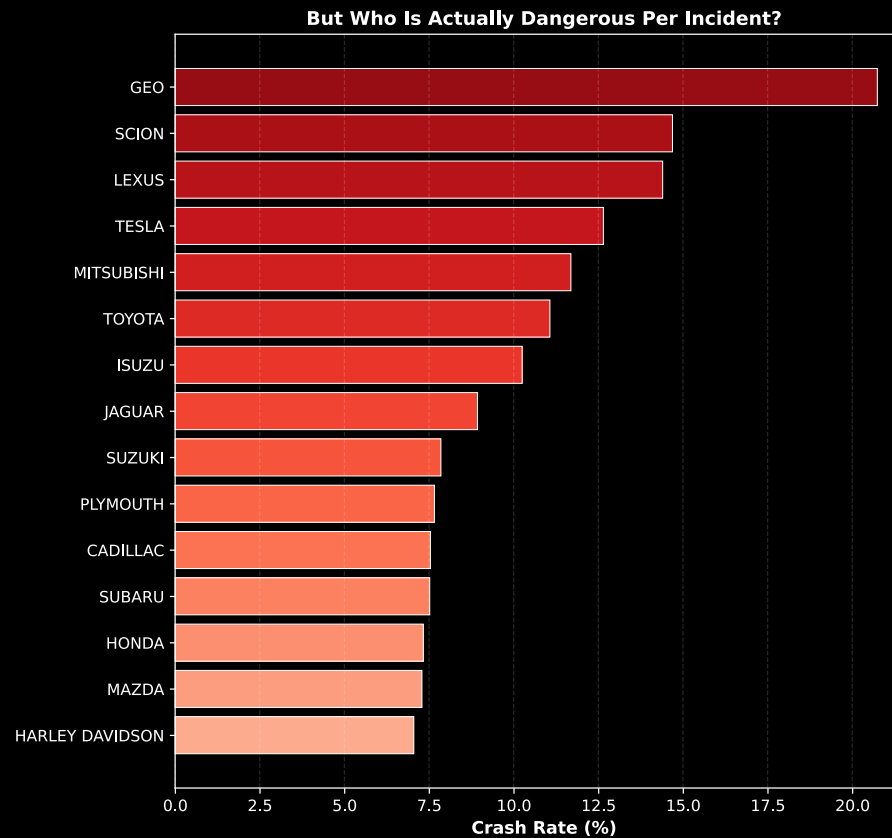
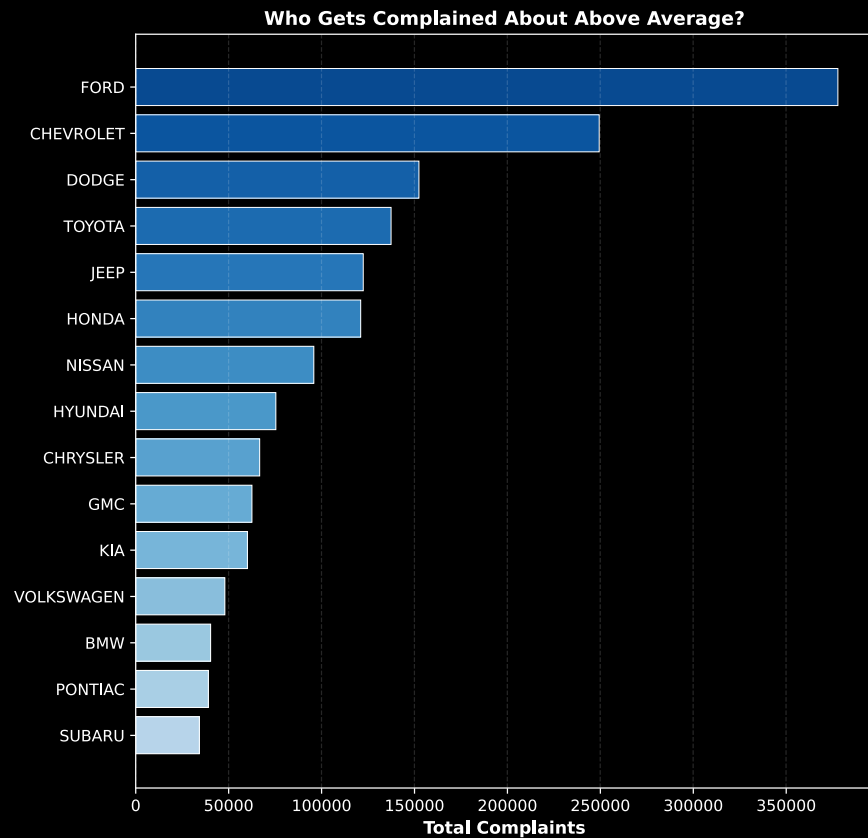
BRAND X COMPONENT X RECALL TYPE



- ▶ Ford's tire and wheels have the highest affected vehicles per campaign but relatively low amount of recall campaigns
- ▶ Conversely, Ford's airbags and electrical systems affect less vehicles per campaign but have more recalls and total vehicles affected

ABOVE-AVERAGE COMPLAINT BENCHMARK

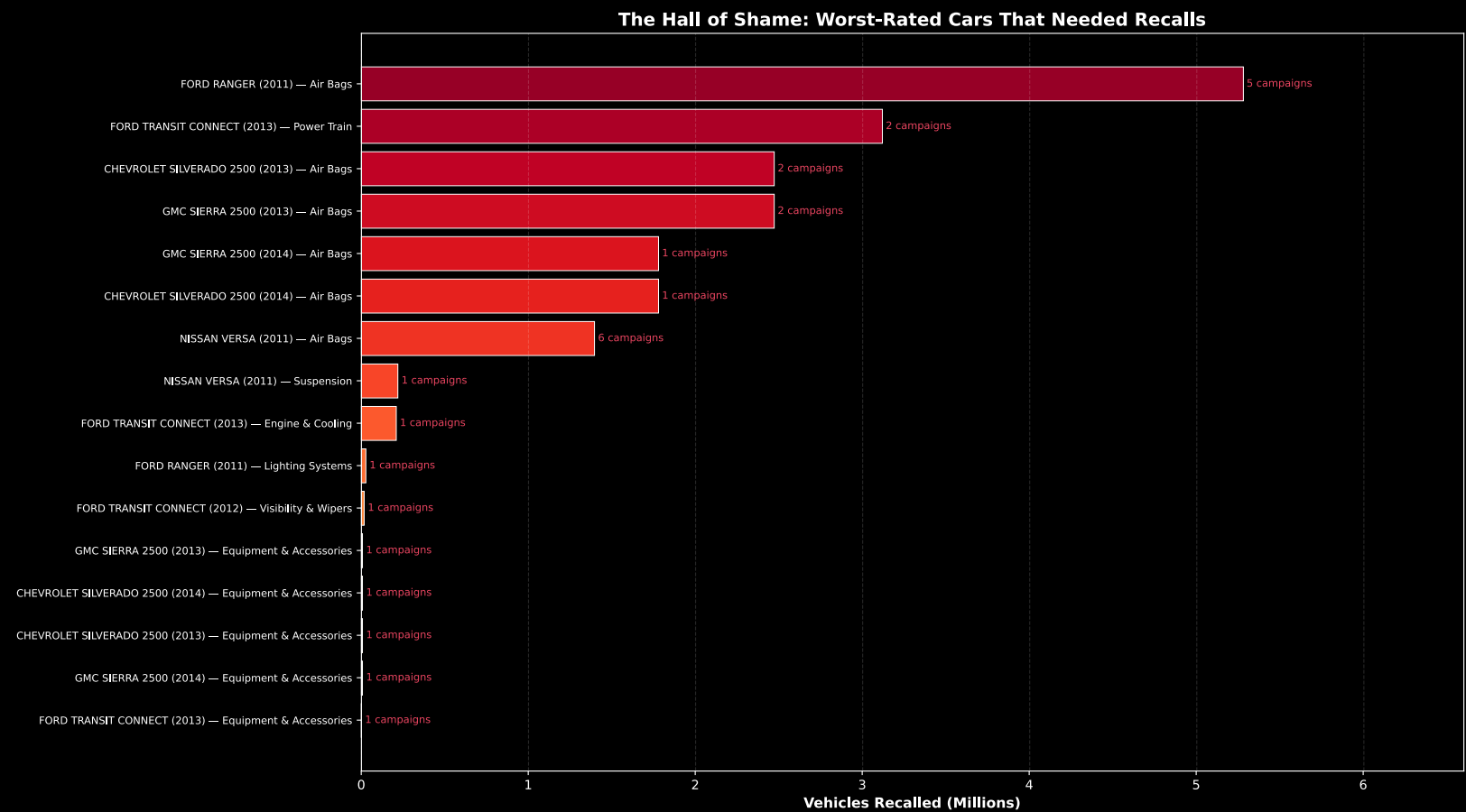
More Complaints = More Dangerous? Not Always.



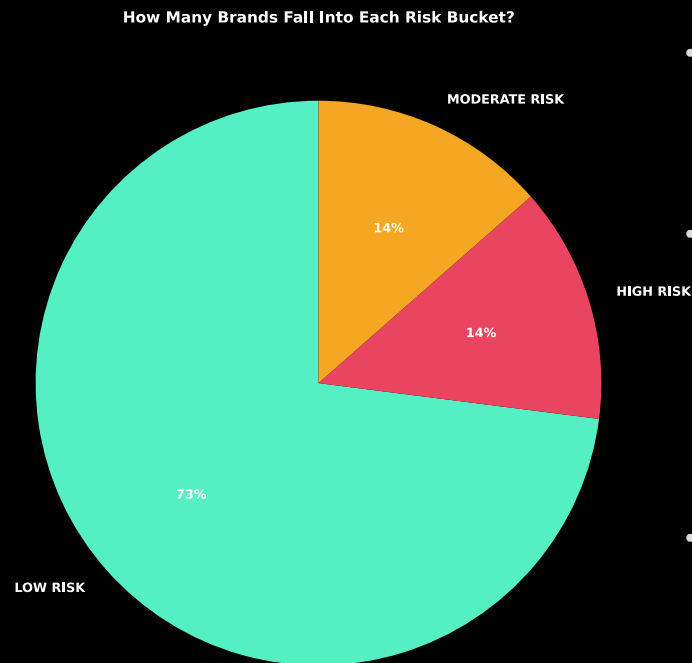
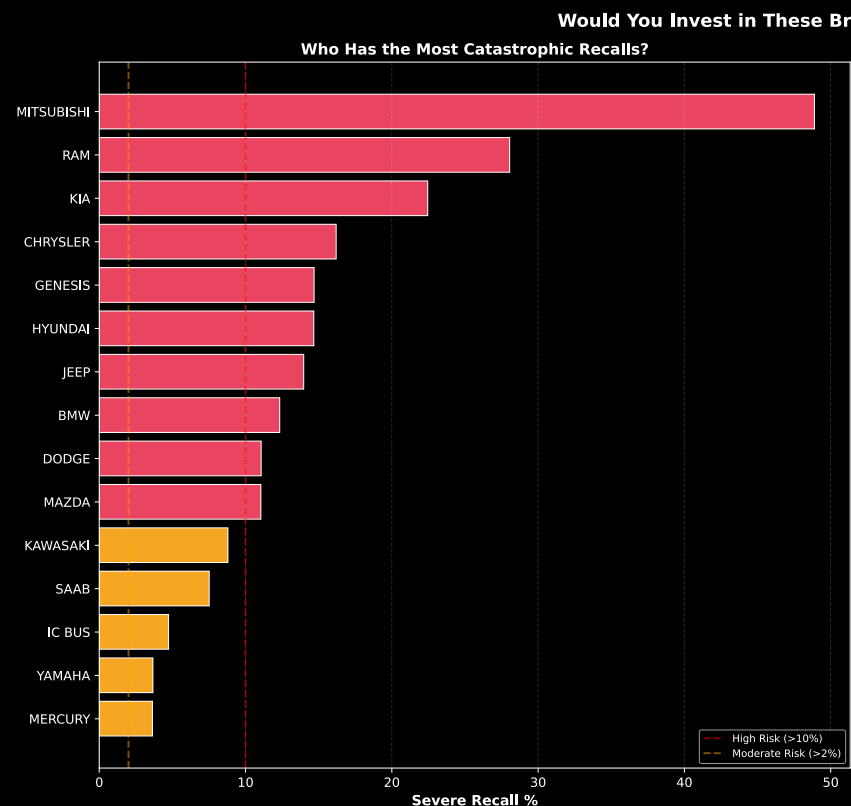
- There is a mismatch between the makes with the most complaints and the makes with the highest crash rates.
- Historical brands like Geo and Scion have low complaint volumes but have high crash rates

HALL OF SHAME - WORST-RATED RECALLED VEHICLES

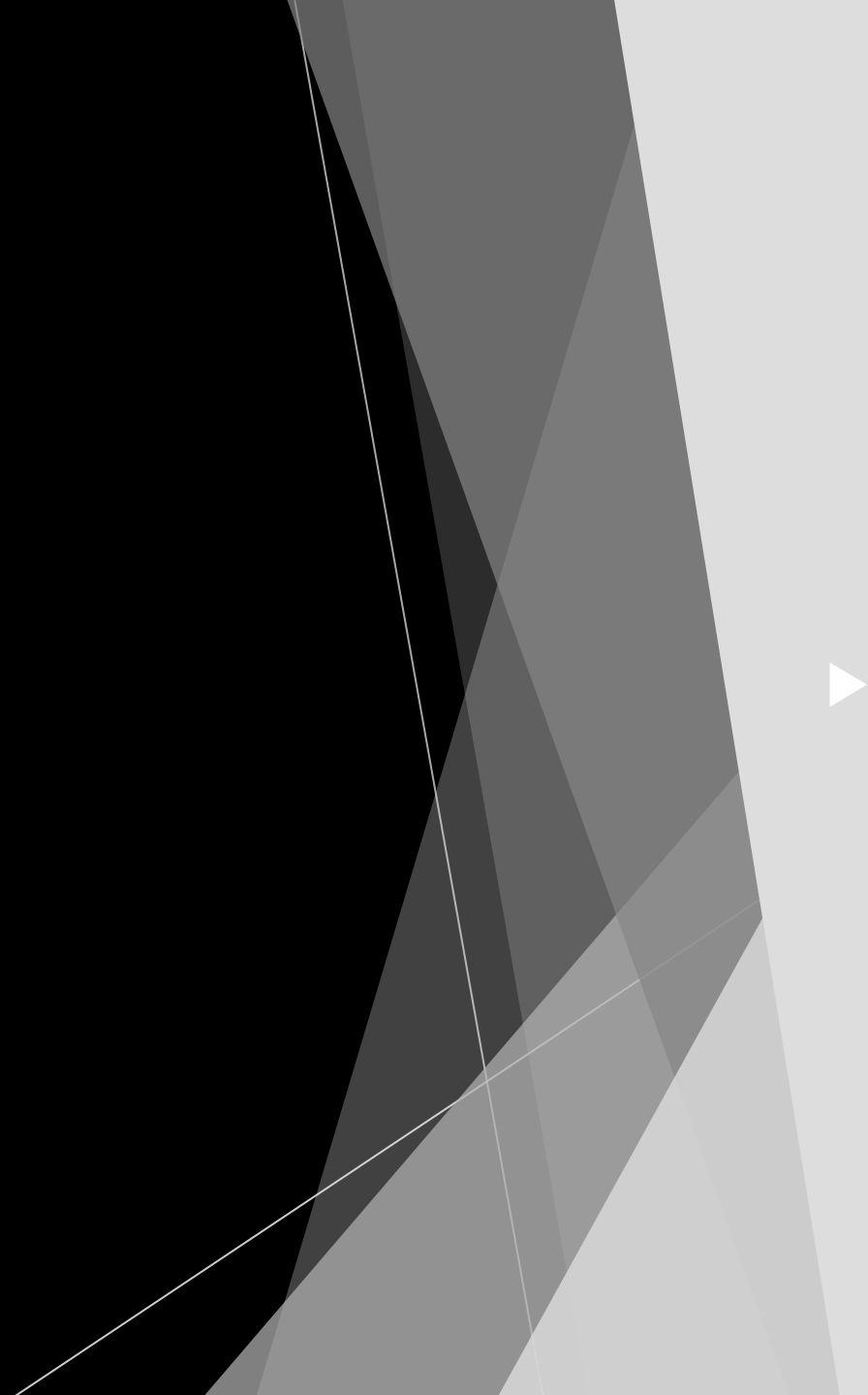
- Vehicles that scored low on crash test ratings (≤ 2 overall stars) also require more recall campaigns
- The same vehicles appear across multiple component categories



BRAND SAFETY RISK TIERS



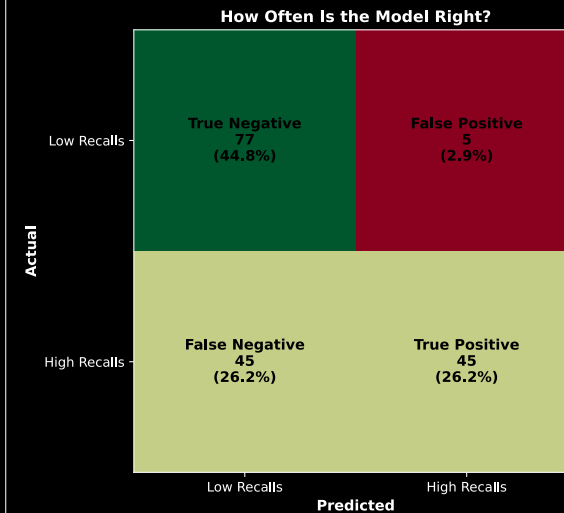
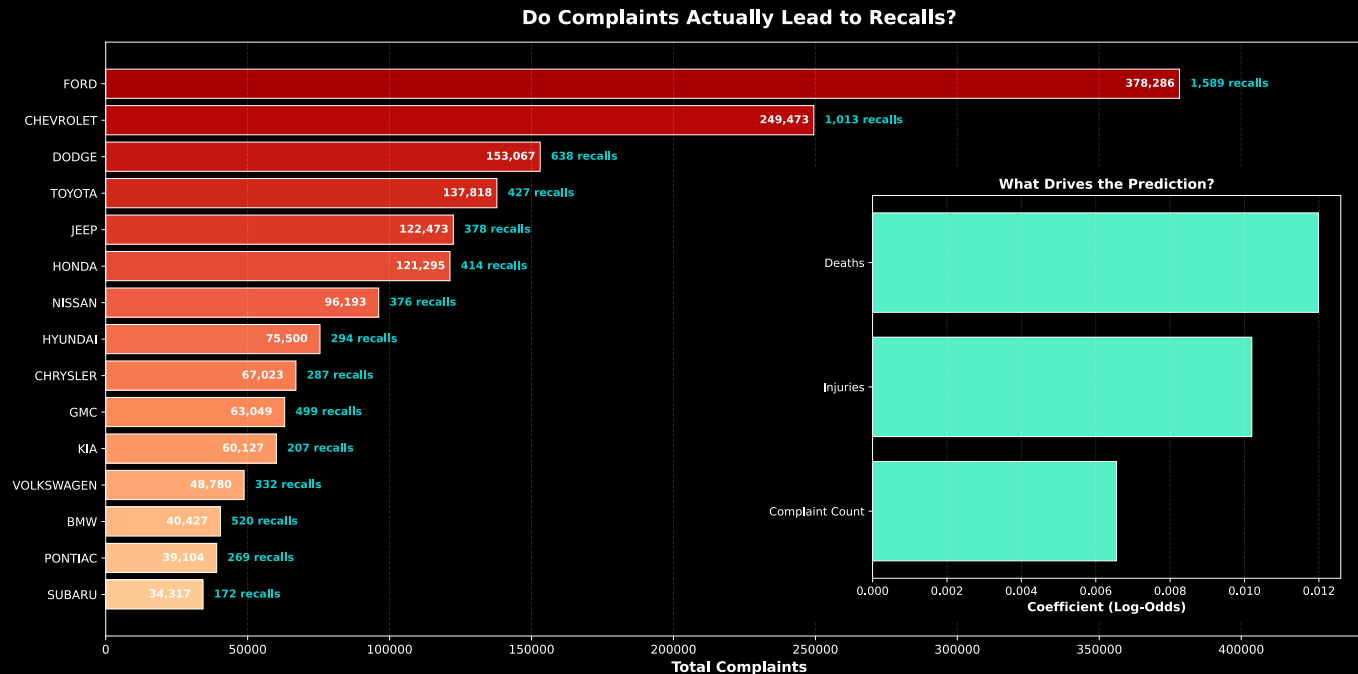
- Mitsubishi leads with the highest percentage of recall volume classified as high risk at 48.9%
- Tesla, Honda, Acura, and Buick had zero severe risk recalls, implying most of their recall volume is software related or routine maintenance-level issues.
- Most brands fall into the low risk bucket, meaning catastrophic recalls are only prevalent in a handful of brands.



PART 7

PREDICTIVE POSSIBILITY

Predicting Major Recalls - Logistic Regression



- Confusion matrix indicates that the model's strength is in identifying low-recall cases correctly (true negative) but is more likely to miss a high-recall case than falsely flag a low recall.
- Deaths and injuries drive the prediction model more than complaints count

► PUTTING IT ALL TOGETHER

What do we consider?

Decision Factor	What This Analysis Shows
Crash test ratings	Necessary but not sufficient — check all four categories, not just the overall star count
Complaint history	Look at per-incident rates (crash %, casualty %), not raw complaint counts
Recall history	Check platform siblings, not just your exact model
Brand risk tier	Avoid brands with high catastrophic recall percentages unless you accept the liability
Safety technology	Brands with OTA-updatable systems carry lower long-term recall risk